



Quiz

Topic: SPL, ERO

1. Given a system of linear equations:

$$\begin{cases} x + 2y + z = 4 \\ 2x + 5y + 3z = 11 \\ -x - y + z = 0 \end{cases}$$

The number of solutions of the system is ?

2. Given a system of linear equations:

$$\begin{cases} x + y + z = 1 \\ 2x + 3y + 4z = 2 \\ x + 2y + (k + 1)z = k \end{cases}$$

The system has infinite solutions if the expiry value ?

3. If the constant k satisfies the following equation

$$\begin{pmatrix} k & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x - 1 \\ y - 1 \end{pmatrix} = \begin{pmatrix} 0 \\ k \end{pmatrix}$$

then the value of $x + y$? (the solution contain k)

4. If $x : y = 5 : 4$, then the values of x and y that satisfy the following equation are

$$\begin{pmatrix} 2 & 10 & 1 \end{pmatrix} \begin{pmatrix} x & y \\ 4 & 5 \\ 30 & 25 \end{pmatrix} \begin{pmatrix} 5 \\ 10 \end{pmatrix} = 1360$$

5. $\begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}$ and a_{ij} is the element of magtrix A^{10} in the i-th row and j-th column, if $a_{21} = p^2 - q$, with p and q are positive integers, then the value $p + q$?

6. if $A = \begin{pmatrix} a & b \\ b & a \end{pmatrix}$, $b \neq 0$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ satisfies $A \cdot A = A + I$, then find b^2

7. If $x = 1, y = -1, z = 2$ is the solution of SPL asf,

$$\begin{pmatrix} a & b & -3 \\ -2 & -b & c \\ a & 3 & -c \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -3 \\ -1 \\ -3 \end{pmatrix}, \text{ find } a^2 - bc =$$

8. If $A = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, then find $(A + B)(A - B) - (A - B)(A + B) =$